

Decoding the Retail Customer

A Shopping Behavior & Subscription Analysis Across 3,900 Transactions

1. Project Overview

This case study digs into 3,900 individual retail transactions spanning several product categories to understand how customers shop, spend, and respond to subscription offers. Rather than just describing what happened, the goal is to turn that raw transaction history into concrete, decision-ready insight — on spending behavior, customer segments, product performance, and subscription engagement — that can directly shape strategy.

2. Dataset Snapshot

Transactions: 3,900 | **Attributes:** 18

- **Who's shopping:** Age, Gender, Location, Subscription Status
- **What they bought:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **How they shop:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

Data quality flag: the Review Rating field arrived with 37 blank entries, which were addressed during cleaning (see below).

3. Cleaning & Exploration in Python

Before any analysis began, the raw file went through a standard cleaning pass in pandas:

- **Loading the data:** the source CSV was read directly into a pandas DataFrame.
- **First look:** `df.info()` and `.describe()` were used to check data types and pull summary statistics for every column.
- **Filling gaps:** the 37 missing Review Rating values were imputed using the median rating within each product category, so the fix stayed consistent with how that category is typically rated.
- **Renaming columns:** headers were converted to snake_case for a cleaner, more consistent schema going into SQL.
- **New fields:** an `age_group` column was built by bucketing customer ages, and a `purchase_frequency_days` column was derived from the existing purchase-frequency text labels.
- **Redundancy check:** `discount_applied` and `promo_code_used` were compared and found to capture almost the same behavior, so `promo_code_used` was dropped.
- **Handing off to SQL:** the cleaned DataFrame was loaded into a MySQL database so the next stage of analysis could run as SQL queries.

Summary statistics produced by `.describe()`:

Stat	Customer ID	Age	Purchase Amt	Review Rating
count	3,900	3,900	3,900	3,863
mean	1,950.5	44.07	59.76	3.75

Stat	Customer ID	Age	Purchase Amt	Review Rating
std	1,125.98	15.21	23.69	0.72
min	1.0	18	20	2.5
25%	975.75	31	39	3.1
50%	1,950.5	44	60	3.8
75%	2,925.25	57	81	4.4
max	3,900	70	100	5.0

Stat	Gender (top)	Category (top)	Location (top)	Subscribed?	Payment (top)
unique	2	4	50	2	6
top / mode	Male	Clothing	Montana	No	PayPal
freq	2,652	1,737	96	2,847	677

Selected fields from the full df.describe() output; Previous Purchases ranged 1–50 (mean 25.4), and 57% of orders did not use a discount.

4. Business Questions Answered in SQL

With the data sitting in MySQL, ten targeted queries were written to answer specific business questions:

4.1 Spending Split by Gender

Total revenue generated by male customers vs. female customers.

Gender	Total Revenue (USD)
Female	\$75,191
Male	\$157,890

4.2 Discount Shoppers Who Still Spend Big

Customers who redeemed a discount but still spent above the overall average purchase amount — 839 customers met this criteria in total. A sample:

Customer ID	Purchase Amount (USD)
2	\$64
3	\$73
4	\$90
7	\$85
9	\$97
12	\$68
13	\$72
16	\$81
20	\$90

Customer ID	Purchase Amount (USD)
22	\$62

Showing 10 of 839 matching customers.

4.3 Highest-Rated Products

The five items with the best average customer review scores.

Item	Avg. Review Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4.4 Spend by Shipping Method

Average purchase amount for Standard vs. Express shipping.

Shipping Type	Avg. Purchase Amount (USD)
Standard	\$58.46
Express	\$60.48

4.5 Subscriber Value Check

How subscribers and non-subscribers compare on spend and total revenue.

Subscribed?	Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

4.6 Products Most Reliant on Discounts

The five items with the highest share of purchases made at a discount.

Item	Share Discounted
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

4.7 New / Returning / Loyal Breakdown

Customers grouped into lifecycle segments based on purchase history.

Segment	Customers
Loyal	3,116
Returning	701
New	83

4.8 Category Leaders

The most-ordered items within each product category.

Rank	Category	Item	Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

4.9 Do Repeat Buyers Subscribe More?

Whether customers with more than 5 prior purchases are more likely to subscribe.

Subscribed?	Repeat Buyers (>5 purchases)
No	2,518
Yes	958

Repeat purchasing alone doesn't appear to be a strong predictor of subscription uptake — the majority of frequent buyers are still not subscribers.

4.10 Revenue Contribution by Age Bracket

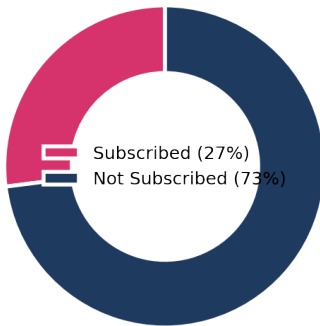
Total revenue generated by each customer age group.

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

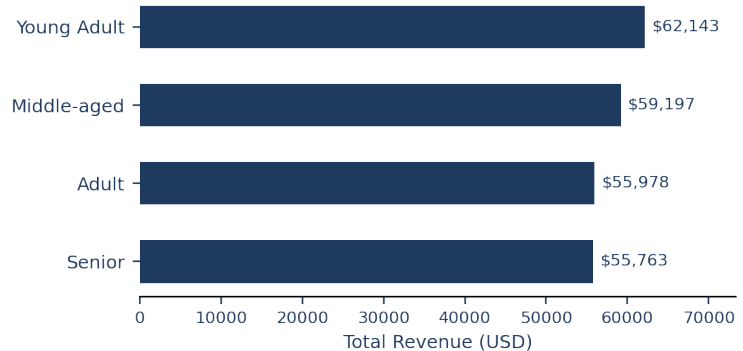
5. Bringing It Together in Power BI

The cleaned dataset and SQL outputs were wired into an interactive Power BI dashboard so non-technical stakeholders could explore the findings themselves. The dashboard opens with three headline KPI cards — **3.9K customers**, an **average purchase amount of \$59.76**, and an **average review rating of 3.75** — backed by slicers for subscription status, gender, category, and shipping type, plus visuals breaking out revenue and sales by both product category and age group.

Customers by Subscription Status



Revenue by Age Group



Two of the dashboard's core visuals, rebuilt here from the underlying query results: the subscription split (27% subscribed) and total revenue by age group.

6. What the Data Suggests We Do Next

- **Grow the subscriber base.** Subscribers currently make up only 27% of customers — surfacing member-only perks more prominently in marketing could move that number.
- **Build a loyalty program.** With 701 customers sitting in the “Returning” tier, a structured rewards program could be the nudge that moves them into “Loyal.”
- **Revisit the discount strategy.** Several products (Hats, Sneakers, Coats) are sold at a discount roughly half the time — worth checking whether that's protecting margin or simply training customers to wait for a sale.
- **Lean into what's already working.** Top-rated items (Gloves, Sandals, Boots) and category leaders (Jewelry, Blouses, Jackets) are natural candidates for featured placement in campaigns and on-site merchandising.
- **Sharpen targeting.** Young Adult and Middle-aged customers, along with Express-shipping users, generate the highest revenue per group — a logical place to concentrate marketing spend.